Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If there are six loops for $1\ \mathrm{m}$ length in transverse mode of Melde's experiment., the no. of loops in longitudinal mode under otherwise identical conditions would be

(A) 3

(B) 6

(C) 12

(D) 8

Q.2

Of the following, the equation of plane progressive wave is

(A) $y = r\sin\omega t$

(B) $y = r\sin(\omega t - kx)$

(C) $y = \frac{a}{\sqrt{r}}\sin(\omega t - kx)$

(D) $y = \frac{a}{r}\sin(\omega t - kr)$

Q.3

At $\mathrm{t} = 0$, the shape of a travelling pulse is given by

$$y(x,0) = \frac{4\times10^{-3}}{8-(X)^2}$$

where x and y are in metres. The wave function for the travelling pulse if the velocity of propagation is $5\ \mathrm{m/s}$ in the x direction is given by

(A) $y(x,t) = \frac{4\times10^{-3}}{8-(x^2-5t)}$

(B) $y(x,t) = \frac{4\times10^{-3}}{8-(x-5t)^2}$

(C) $y(x,t) = \frac{4\times10^{-3}}{8-(x+5t)^2}$

(D) $y(x,t) = \frac{4\times10^{-3}}{8-(x^2+5t)}$

Q.4

@IITJEE_Advanced

A plane progressive simple harmonic sound wave of angular frequency $680 \mathrm{rad/s}$ moves with speed $340 \mathrm{~m/s}$ in the direction which makes equal angle with each x, y and z -axis. The phase difference $\left(\phi_1-\phi_2\right)$ between the oscillations of the particle in the medium located at the positions $(\sqrt{3}, \sqrt{3}, \sqrt{3})$ and $(2\sqrt{3},, 2\sqrt{3}, 2\sqrt{3})$ is (assume $\cos\theta > 0$)

(A) 2 radian

(B) 3 radian

(C) 4 radian

(D) 6 radian

Q.5

The amplitude of a wave disturbance propagating in the positive x -direction is given by $y=\frac{1}{1+x^2}$ at $t=0$ and $y=\frac{1}{2+x^2-2x}$ at $t=2s$, where x and y are in meter. Assuming that the shape of the wave disturbance does not change during the propagation, the speed of the wave is

(A) $0.5 \mathrm{~m/s}$

(B) $1 \mathrm{~m/s}$

(C) $1.5 \mathrm{~m/s}$

(D) $2 \mathrm{~m/s}$

Q.6

The wave described by $\mathrm{y}=0.25 \sin (10 \pi \mathrm{x}-2 \pi \mathrm{t})$,
where x and y are in meters and t in seconds, is a wave travelling along the:

(A) $-$ ve x direction with frequency $1 \mathrm{~Hz}$.

(B) +ve x direction with frequency $\pi \mathrm{Hz}$ and wavelength $\lambda=0.2 \mathrm{~m}$

(C) +ve x direction with frequency $1 \mathrm{~Hz}$ and wavelength $\lambda=0.2 \mathrm{~m}$

(D) $-$ ve x direction with amplitude $0.25 \mathrm{~m}$ and wavelength $\lambda=0.2 \mathrm{~m}$

Q.7

When a longitudinal wave propagates through a medium, the particles of the medium execute simple harmonic oscillations about their mean positions. These oscillations of a particle are characterised by an invariant

(A) kinetic energy

(B) potential energy

(C) sum of kinetic energy and potential energy

(D) difference between kinetic energy and potential energy

@IITJEE_Advanced

Q.8

In a transverse wave the distance between a crest and neighbouring trough at the same instant is 4.0 cm and the distance between a crest and trough at the same place is $1.0\ \mathrm{cm}$. The next crest appears at the same place after a time interval of $0.4\ \mathrm{s}$. The maximum speed of the vibrating particles in the medium is :

(A) $\frac{3\pi}{2}\ \mathrm{cm/s}$

(B) $\frac{5\pi}{2}\ \mathrm{cm/s}$

(C) $\frac{\pi}{2}\ \mathrm{cm/s}$

(D) $2\pi \mathrm{cm/s}$

Q.9

In a plane progressive harmonic wave particle speed is always less than the wave speed if

(A) amplitude of wave is less than $\frac{\lambda}{2\pi}$

(B) amplitude of wave is greater than $\frac{\lambda}{2\pi}$

(C) amplitude of wave is less than λ

(D) amplitude of wave is greater than $\frac{\lambda}{\pi}$

Q.10

Equation of a progressive wave is given by $y = 4\sin\left[\pi\left(\frac{t}{5} - \frac{x}{9}\right) + \frac{\pi}{6}\right]$
Then which of the following is correct?

(A) $\mathrm{v} = 5\ \mathrm{cm}$

(B) $\lambda = 18\ \mathrm{cm}$

(C) $\mathrm{a} = 0.04\ \mathrm{cm}$

(D) $\mathrm{f} = 50\ \mathrm{Hz}$

Q.11

When a sound wave of frequency $300\ \mathrm{Hz}$ passes through a medium, the maximum displacement of a particle of the medium is $0.1\ \mathrm{cm}$. The maximum velocity of the particle is equal to

(A) $60\pi \mathrm{cms}^{-1}$

(B) $30\pi \mathrm{cms}^{-1}$

(C) $30 \mathrm{cms}^{-1}$

(D) $60 \mathrm{cms}^{-1}$

Q.12

@IITJEE_Advanced

If the length of a stretched string is shortened by 40% and the tension increased by 44% then the ratio of the final and initial fundamental frequencies is

(A) $3:4$

(B) $4:3$

(C) $1:3$

(D) $2:1$

Q.13

Two strings A and B, made of same material, are stretched by same tension. The radius of string A is double of radius of B. A transverse wave travels on A with speed v_A and on B with speed $\mathrm{v_B}$. The ratio $\mathrm{v_A/v_B}$ is

(A) $1/2$

(B) 2

(C) $1/4$

(D) 4

Q.14

The equation of a wave on a string of linear mass density $0.04\ \mathrm{kg\ m^{-1}}$ is given by $y = 0.02(m)\sin\left[2\pi\left(\frac{t}{0.04(s)} - \frac{x}{0.50(m)}\right)\right]$
The tension in the string is

(A) $4.0\ \mathrm{N}$

(B) $12.5\ \mathrm{N}$

(C) $0.5\ \mathrm{N}$

(D) $6.25\ \mathrm{N}$

Q.15

A stretched wire $60\ \mathrm{cm}$ long is vibrating with its fundamental frequency of $256\ \mathrm{Hz}$. If the length of the wire is decreased to $15\ \mathrm{cm}$ and the tension remains the same. Then the fundamental freuqency of the vibration of the wire will be

(A) 1024

(B) 572

(C) 256

(D) 64

Q.16

@IITJEE_Advanced

A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2 the ratio λ_2/λ_1 is

(A) $\sqrt{\frac{m_1}{m_2}}$

(B) $\sqrt{\frac{m_1+m_2}{m_2}}$

(C) $\sqrt{\frac{\mathrm{m}_2}{\mathrm{m}_1}}$

(D) $\sqrt{\frac{\mathrm{m}_1+\mathrm{m}_2}{\mathrm{m}_1}}$

Q.17

The displacement wave is $\xi = 2.0\cos(10t - 0.008x + 0.50)$, where ξ and x are in cm and is in s. The phase difference between two particles separated by a distance of $50\ \mathrm{cm}$ is

(A) $0.4\mathrm{rad}$

(B) $0.04\mathrm{rad}$

(C) $0.004\mathrm{rad}$

(D) none of these

Q.18

When a wave travel in a medium, the particle displacement is given by the equation $\mathrm{y} = \mathrm{a}\sin 2\pi(\mathrm{bt} - \mathrm{cx})$ where a, b and
c are constants. The maximum particle velocity will be twice the wave velocity if

(A) $\mathrm{c} = \frac{1}{\pi \mathrm{a}}$

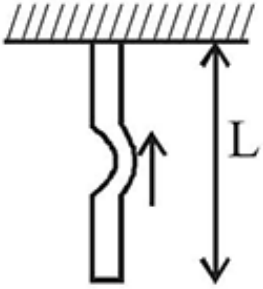
(B) $\mathrm{c} = \pi \mathrm{a}$

(C) $\mathrm{b} = \mathrm{ac}$

(D) $\mathrm{b} = \frac{1}{\mathrm{ac}}$

Q.19

A thick uniform rope of length L is hanging from a rigid support. A transverse wave of wavelength λ_0 is set up at the middle of rope as shown in figure. The wavelength of the wave as it reaches to the topmost point is



@IITJEE_Advanced

(A) $2\lambda_0$

(B) $\sqrt{2}\lambda_0$

(C) $\frac{\lambda_0}{\sqrt{2}}$

(D) λ_0

Q.20

An object of specific gravity ρ is hung from a thin steel wire. The fundamental frequency for transverse standing waves in the wire is $300\ \text{Hz}$. The object is immersed in water so that one half of its volume is submerged. The new fundamental frequency in Hz is

(A) $300\left(\frac{2\rho-1}{2\rho}\right)^{1/2}$

(B) $300\left(\frac{2\rho}{2\rho-1}\right)^{1/2}$

(C) $300\left(\frac{2\rho}{2\rho-1}\right)$

(D) $300\left(\frac{2\rho-1}{2\rho}\right)$

Answers & Solutions

Q.1 Answer:

3

Solution:

One cycle of up and down vibration for transverse waves on the string is two cycles of string tension increase and decrease. The tension is maximum both at the loops' maximum up position and again at maximum down position. Therefore, in longitudinal drive mode, since the string tension increases and decreases once per tuning fork vibration, it takes one tuning fork vibration to move the string loop to maximum up position and one to move it to maximum down position. This is two tuning fork vibrations for one up and down string vibration, so the tuning fork frequency is half the string frequency.
Hence, the number of loops in longitudinal mode(tuning fork) is half of that of transverse mode(string).

Q.2 Answer:

$y = r\sin(\omega t - kx)$

Solution:

The position of such a wave changes in two dimensional plane with time.

Q.3 Answer:

@IITJEE_Advanced

$y(x,t)=\frac{4\times10^{-3}}{8-(x-5t)^2}$

Solution:

$y(x,t)=f(x-vt)$

$y=(x,0)=\frac{4\times10^{-3}}{8-x^2}$

For a travelling wave in the x -direction

$y(x,t)=\frac{4\times10^{-3}}{8-(x-5t)^2}$

Q.4 Answer:

6 radian

Solution:

$y(\vec{r},t)=A\sin(\omega t-\vec{k}\cdot\vec{r})$

$\hat{k}=\frac{2\pi}{\lambda}(\cos\alpha\hat{i}+\cos\beta\hat{j}+\cos\gamma\hat{k})=\frac{k}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$

$\phi_1-\phi_2=2(3)=6\text{rad}$

Q.5 Answer:

$0.5\ \text{m/s}$

Solution:

At $t=0, y=\frac{1}{1+x^2}$ **or** $x=\sqrt{\frac{1-y}{y}}=x_1$

At $t=2s,\quad y=\frac{1}{2+x^2-2x}=\frac{1}{1+(x-1)^2}$

or $(x-1)^2=\frac{1-y}{y}$ **or** $x=1+\sqrt{\frac{1-y}{y}}=x_2$

Speed of the wave

$v=\frac{\Delta x}{\Delta t}=\frac{x_2-x_1}{t_2-t_1}=\frac{1}{2-0}=0.5\ \text{m/s}$

Q.6 Answer:

+ve x direction with frequency $1\ \text{Hz}$ and wavelength $\lambda=0.2\ \text{m}$

Solution:

$y=0.25\sin(10\pi x-2\pi t)$

Comparing this equation with the standard wave equation $\text{y}=\text{asin}(\text{kx}-\omega\text{t})$

We get, $\text{k}=10\pi\Rightarrow\frac{2\pi}{\lambda}=10\pi\Rightarrow\lambda=0.2\ \text{m}$

And $\omega=2\pi$ **or,** $2\pi v=2\pi\Rightarrow v=1\ \text{Hz}$

The sign inside the bracket is negative, hence the wave travels in $+$ **ve** $x-$ **directior**

Q.7 Answer:

sum of kinetic energy and potential energy

Solution:

The sum of their KE **and** PE **is a constant.**

@IITJEE_Advanced

Q.8 Answer:

$\frac{5\pi}{2}\ \mathrm{cm/s}$

Solution:

Transverse wave equation

$\mathrm{y} = \mathrm{a}\sin(\omega \mathrm{t} + \mathrm{kx})$

Next crest appears at time interval of $0.4\ \mathrm{s}$

So, the time period $\mathrm{T} = 0.4$

$\therefore$ **Angular Frequency** $\omega = \frac{2\pi}{\mathrm{T}} = \frac{2\pi}{0.4}$

$\therefore$ **Speed of the particle**

$v = \frac{\mathrm{dy}}{\mathrm{dt}} = \mathrm{a}\omega\sin(\omega \mathrm{t} + \mathrm{kx})$

$v_{\max} = \mathrm{a}\omega$

$\mathrm{a} = 1/2\ \mathrm{cm}$

$= \frac{1}{2} \times \frac{2\pi}{0.4} = \frac{2\pi}{8} \times 10$

$= \frac{5\pi}{2}\ \mathrm{cm/s}$

Q.9 Answer:

amplitude of wave is less than $\frac{\lambda}{2\pi}$

Solution:

Let, particle speed is $\mathrm{v_p}$ **and wave speed is** v

So, according the given question $\mathrm{v_p} < \mathrm{v}$

Let, equation of wave is $\mathrm{y(x)} = \mathrm{A}\sin(\omega \mathrm{t} - \mathrm{kx})$

Now, $\mathrm{v} = \frac{\omega}{\mathrm{k}}$

The maximum particle speed is $(\mathrm{v_p})_{\mathrm{m}} = \mathrm{A}\omega$

So, paricle speed is always less than wave speed if $(\mathrm{v_p})_{\mathrm{m}} < \mathrm{v}$

Or, $\mathrm{A}\omega < \frac{\omega}{\mathrm{k}}$

Or, $\mathrm{A} < \frac{1}{\mathrm{k}}$

Or, $\mathrm{A} < \frac{\lambda}{2\pi}$ [**Also,** $k = \frac{2\pi}{\lambda}$]

The correct answer is Option (a) **: Amplitude of wave is less than** $\frac{\lambda}{2\pi}$

Q.10 Answer:

$\lambda = 18\ \mathrm{cm}$

Solution:

The standard equation of a progressive wave is $\mathrm{y} = \mathrm{a}\sin\left[2\pi\left(\frac{\mathrm{t}}{\mathrm{T}} - \frac{\mathrm{x}}{\lambda}\right) + \phi\right]$

The given equation can be written as $y = 4\sin\left[2\pi\left(\frac{t}{10} - \frac{x}{18}\right) + \frac{\pi}{6}\right]$

$\therefore \mathrm{a} = 4\ \mathrm{cm},\ \mathrm{T} = 10\ \mathrm{s}, \lambda = 18\ \mathrm{cm}$ **and** $\phi = \pi/6$

Hence, (b) is correct.

Q.11 Answer:

$60\pi \mathrm{cms}^{-1}$

@IITJEE_Advanced

Solution:

$$V_{\max} = a\omega = 0.1 \times 2\pi \times 300 = 60\pi \text{cms}^{-1}$$

Q.12 Answer:

$2 : 1$

Solution:

Length is shortened by 40%

$\Rightarrow$ length $l_2 = \frac{3}{5}$

Tension is increased by 44%

$\Rightarrow$ **Tension** $\mathrm{T}_2 = \frac{144}{100}\ \mathrm{T}_1$

$\therefore \mathrm{f} = \frac{1}{2\mathrm{l}}\sqrt{\frac{\mathrm{T}}{\mu}}$

$\Rightarrow \frac{\mathrm{f}_1}{\mathrm{f}_2} = \frac{\mathrm{l}_2}{\mathrm{l}_1} \times \sqrt{\frac{\mathrm{T}_2}{\mathrm{T}_2}}$

$= \frac{3}{5} \times \frac{10}{12}$

$= \frac{1}{2}$

$\therefore$ **final : initial** $= 2 : 1$

Q.13 Answer:

$1/2$

Solution:

Velocity of a transverse wave in a stretched string $\mathrm{v} = \sqrt{\frac{\mathrm{T}}{\mu}}$

where T **is the tension of the string and** μ **is mass per unit length of the string.**

$\mu = \pi r^2 \times \rho$

Where r **is the radius or the string and** ρ **is the density of the material of the string.**

$\therefore \mathrm{v} = \frac{1}{\mathrm{r}}\sqrt{\frac{\mathrm{T}}{\pi\rho}}$

Since T, ρ **are constant**

$\therefore \mathrm{v} \propto \frac{1}{\mathrm{r}}$

$\frac{\mathrm{v_A}}{\mathrm{v_B}} = \frac{\mathrm{r_B}}{\mathrm{r_A}} = \left(\frac{\mathrm{r_B}}{2\mathrm{r_B}}\right) = \frac{1}{2}.$

Q.14 Answer:

6.25 N

Solution:

Equation of wave is

$$\mathrm{y} = (0.02\ \mathrm{m}) \sin\left[2\pi\left(\frac{\mathrm{t}}{0.04(\mathrm{\ s})} - \frac{\mathrm{x}}{0.5(\mathrm{\ m})}\right)\right]\}$$

wave velocity $= \frac{2\pi/0.04}{2\pi/0.5} = \frac{50}{4}\ \mathrm{m/s}$

$\therefore \quad \mathrm{T} = \mu \mathrm{v}^2 = \frac{0.04}{100} \times \frac{2500}{16} = 6.25$

@IITJEE_Advanced

Q.15 Answer:

1024

Solution:

We've been given that a stretched wire has a fundamental note of frequency 256 Hz when it is 50 cm long and we want to find out the fundamental frequency when the length is cut down to 15 cm.

Using the formula of the frequency of waves on a string, we can write

$f = \frac{v}{2L}$

Where the velocity of a wave is given as

$v = \sqrt{\frac{T}{\mu}}$

Since we've been told the tension in the string remains constant, so will the velocity of the waves travelling on the string will also remain constant. So the frequency will only be affected by the change in length and since

$f \propto \frac{1}{L}$**, we have:**

$\frac{f_1}{f_2} = \frac{L_2}{L_1}$

Substituting the value of $f_1 = 256$ Hz, $L_1 = 60$ cm **and** $L_2 = 15$ cm**, we get**

$\frac{256}{f_2} = \frac{15}{60}$

Solving for f_2**, we get**

$f_2 = 1024Hz$ **which corresponds to option (d).**

Q.16 Answer:

$\sqrt{\frac{m_1+m_2}{m_2}}$

Solution:

$\mathrm{T_1 = m_2 g}$

$\mathrm{T_2 = (m_1 + m_2)\, g}$

Velocity $\propto \sqrt{T}$

$\Rightarrow \lambda \propto \sqrt{\mathrm{T}}$

$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{\sqrt{T_1}}{\sqrt{T_2}}$

$\Rightarrow \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{\mathrm{m_1+m_2}}{\mathrm{m_2}}}$

Q.17 Answer:

$0.4\mathrm{rad}$

Solution:

phase $diff - k\Delta x = 0.008 \times 50 = 0.4$ **rad**

Q.18 Answer:

$\mathrm{c} = \frac{1}{\pi \mathrm{a}}$

@IITJEE_Advanced

Solution:

Equation of the harmonic progressive wave given by:

$y = a \sin 2\pi(bt - cx)$

Here $v = b\, k = \frac{2\pi}{\lambda} = 2\pi c$ **take,** $\frac{1}{\lambda} = c$

$\therefore$ **Velocity of the wave** $= \cup\lambda = b\frac{1}{c} = \frac{b}{c}$

$\frac{dy}{dt} = a2\pi b \cos 2\pi(bt - cx) = a\omega \cos(\omega t - kx)$

Maximum particle velocity $= a\omega = a2\pi b = 2\pi ab$

given this is $2 \times \frac{b}{c}$ **i.e.** $2\pi a = \frac{2}{c}$ **or** $c = \frac{1}{\pi a}$

Q.19 Answer:

$\sqrt{2}\lambda_0$

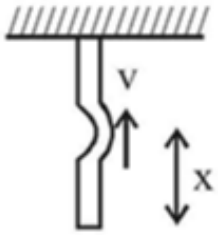
Solution:

Speed of pulse at a distance x

from bottom, $\mathrm{v} = \sqrt{\mathrm{gx}}$

While traveling from mid point $\left\{ \mathrm{v} \right.$

to the top, frequency remains unchanged. $\frac{\mathrm{v}_1}{\lambda_1} = \frac{\mathrm{v}_2}{\lambda_2} \Rightarrow \frac{\sqrt{\mathrm{g(L/2)}}}{\lambda_0} = \frac{\sqrt{\mathrm{gL}}}{\lambda_2} \Rightarrow \lambda_2 = \sqrt{2}\lambda_0$



Q.20 Answer:

$300\left(\frac{2\rho - 1}{2\rho}\right)^{1/2}$

Solution:

In air : $T = \mathrm{mg} = \rho \mathrm{Vg}$

In water : $T = mg -$ **upthrust** $= V\rho g - \frac{V}{2}\rho_\omega g = \frac{Vg}{2}(2\rho - \rho_\omega)$

$\therefore \quad f' = \frac{1}{2\ell}\sqrt{\frac{\frac{Vg}{2}(2\rho - \rho_\omega)}{m}} = \frac{1}{2\ell}\sqrt{\frac{Vg}{m}}\sqrt{\frac{(2\rho - \rho_\omega)}{2\rho}}$

$\frac{f'}{f} = \sqrt{\frac{2\rho - \rho_\omega}{2\rho}} \quad f' = f\left(\frac{2\rho - \rho_\omega}{2\rho}\right)^{1/2}$

@IITJEE_Advanced